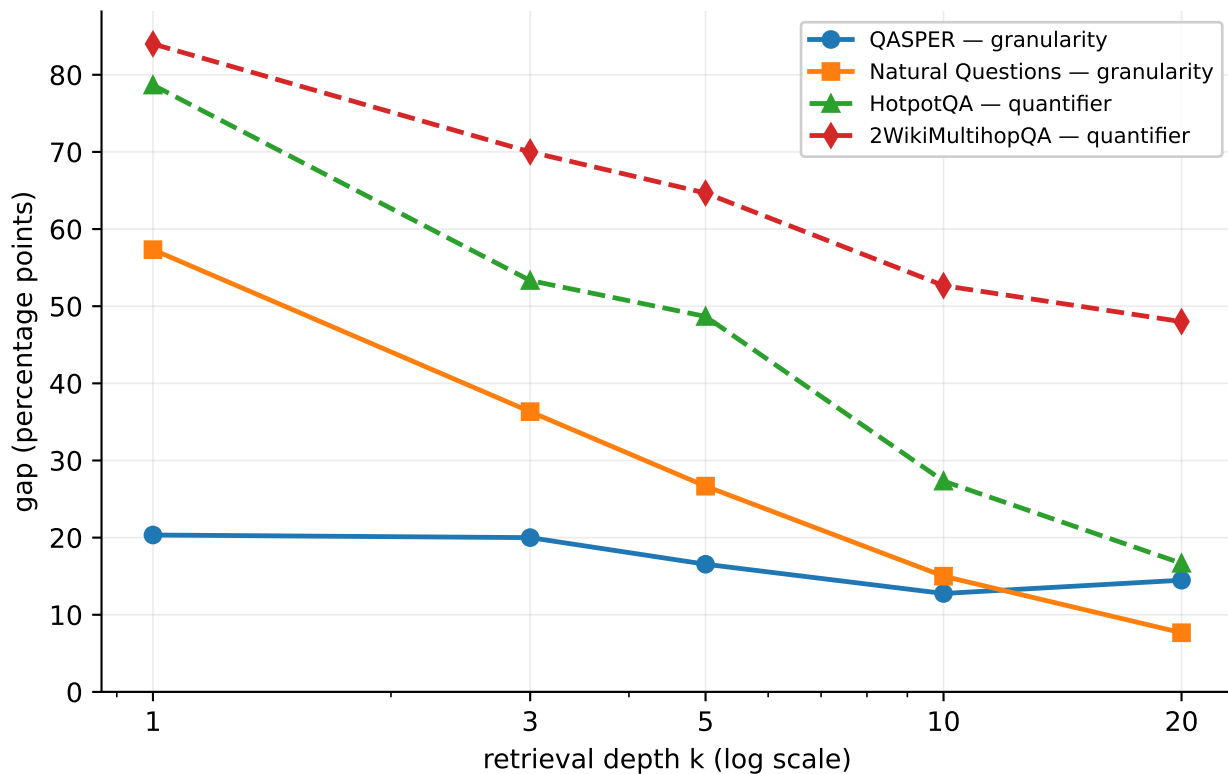


Both effects persist across retrieval depth — with very different slopes



Every depth is retrieved natively — the query is re-issued at each k rather than one deep ranking being truncated. Corpus, chunking (256/32) and embedder (MiniLM) are held constant; only k varies. Solid: granularity (B→C). Dashed: quantifier (A→B). Every point is significant (exact McNemar, $p < 1e-6$). The $k=1$ multi-hop points are definitional — a 2-hop question cannot hold both required documents in one slot — and carry no evidential weight.